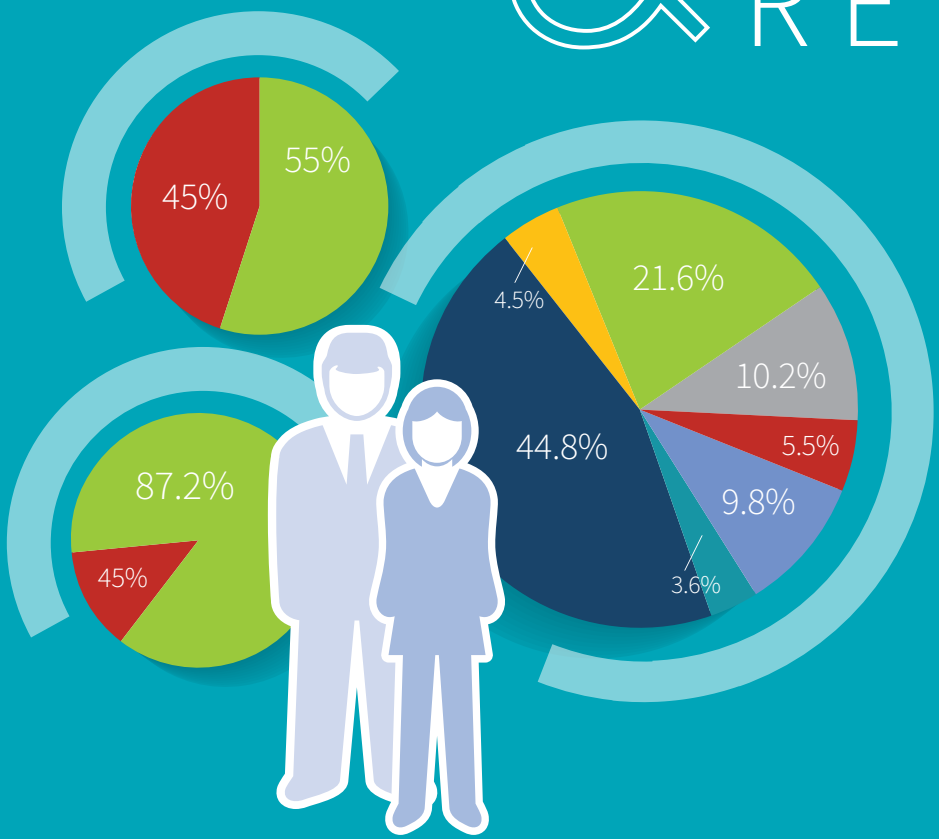
Hydraulics & Pneumatics®

2017 SALARY & CAREER REPORT



LEAH SCULLY | Associate Editor

2017 *Hydraulics & Pneumatics*

ANNUAL SALARY & CAREER REPORT:

Engineers Weigh In on Education and the State of the Workforce

Based on our survey of more than 1,000 fluid power engineers, more than 90% of respondents said they were satisfied in their current positions, and a similar percentage would recommend the profession to young people. In addition, the engineers have lots of strong opinions about formal education, team-building, and staying updated on relevant topics to further their engineering careers.

When asked whether a shortage of fluid power talent exists, some respondents suggested that colleges and institutions should increase students' lab time and offer more opportunities for hands-on work to improve their success in STEM careers. Other responses noted the importance of learning how to apply engineering principles and industry standards during formal education. Apprenticeships and internships were also highlighted as key opportunities for students to build skills and relationships that could jumpstart their careers. About half of the respondents' companies provide internship positions to college and high-school students.

"Engineering schools need to give students more opportunities to work on hand-on projects," one respondent said. "The students need to learn to apply theory and math in reality to develop products or improve an apparatus or machines. Most curriculums don't have enough lab time."

When prompted whether companies should be expected to provide comprehensive in-house training to entry-level workers, many respondents said that they should, while suggesting that formal education programs should also share responsibility by preparing students with application-based training based on their preferred field of study. Meanwhile, about half of the respondents' companies provide mentoring programs for workers, while less than 10% use training consultants to train engineers before they start the job. This could mean that company employees are helping train entry-level workers on the job.

But survey results also show that entry-levels are indeed bringing new skills to their companies, including those in CAD/CAM modeling and simulation, 3D printing, and robotics and computer programming. One commenter noted that many entry-level workers are using more "social media techniques to collaborate and share information," while another recognized an "inherent intuition (among entry-level workers) for finding information nested deep in software programs."

More than 60% of respondents said that their companies are hiring specialists for emerging technologies including electromechanics and electronic systems, themes in mechanical design, controls, automation, and predictive maintenance. Respondents see a potential for job growth in automation, machine monitoring and controls, and data gathering and analysis within the next five to 10 years. hp

THE TYPICAL ENGINEER





Average Age	
25-29	2.3%
30-34	3.1%
35-39	4.0%
40-44	4.9%
45-49	8.6%
50-54	18.9%
55-59	19.1%
60 or older	42.0%

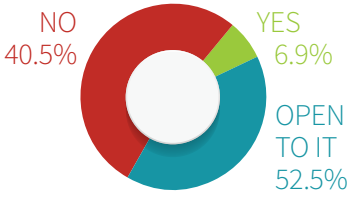
COMPENSATION



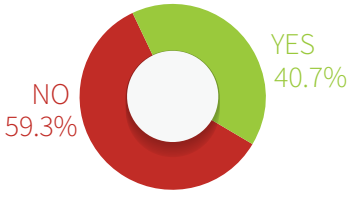
- AVG SALARY $93,345
- AVG BONUS $5,454
- AVG STOCK OPTIONS $5,315
- AVG OTHER SOURCES $7,174

EMPLOYMENT OUTLOOK

Actively seeking a new position



Focused on employee retention



Contacted by recruiter 58.2%



YEARS IN THE PROFESSION

Less than 1 year	2.9%
1-4 years	4.4%
5-9 years	6.1%
10-14 years	7.1%
15-19 years	8.2%
20-24 years	12.0%
25-29 years	12.9%
30-34 years	14.4%
35-39 years	12.9%
40 years or more	19.2%

WORK LOCATION

Ohio	9.7%
Michigan	5.7%
Texas	6.1%
Illinois	4.8%
Wisconsin	5.8%
New York	3.1%
Pennsylvania	6.1%
Washington	3.3%
Indiana	3.1%

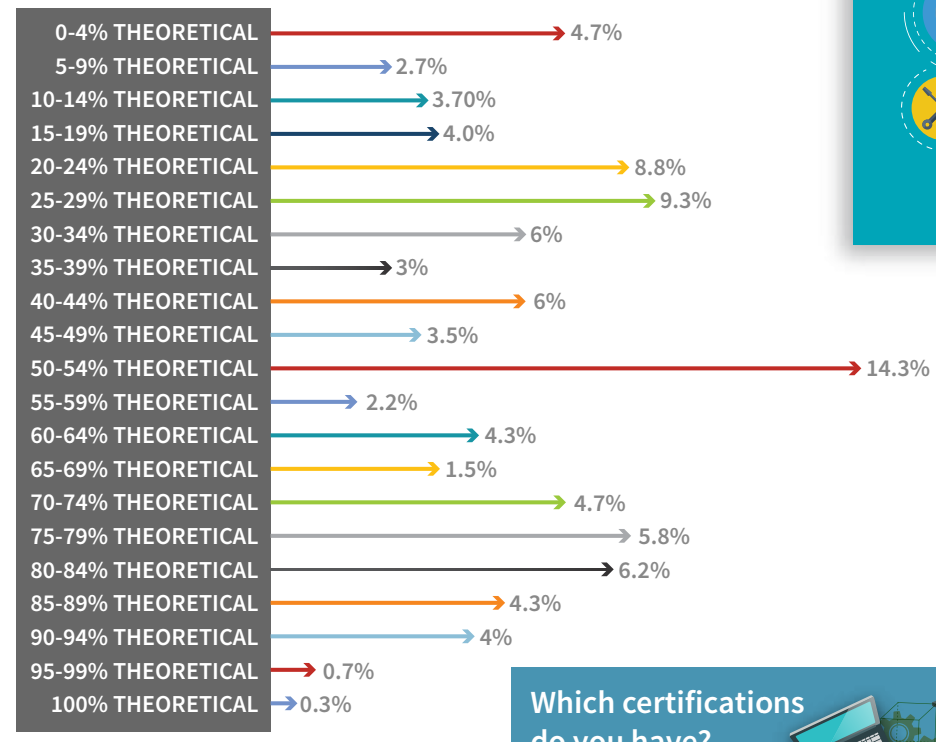
YEARS AT PRESENT COMPANY

Currently unemployed	1.0%
Less than 1 year	3.9%
1-4 years	18.9%
5-9 years	17.3%
10-14 years	15.6%
15-19 years	8.5%
20-24 years	9.9%
25-29 years	7.4%
30-34 years	7.9%
35-39 years	4.6%
40 years or more	5.0%

YOUR EDUCATION



APPROXIMATELY WHAT PERCENTAGE OF THE KNOWLEDGE YOU RECEIVED IN YOUR EDUCATIONAL TRAINING IS THEORETICAL ENGINEERING (PUTTING TOGETHER A FUNDAMENTAL KNOWLEDGE BASE) VS. PRACTICAL APPLICATIONS (TAKING THAT KNOWLEDGE AND APPLYING IT TO A REAL-WORLD PROBLEM)?



AVERAGE **46.78%**

What do you think are the most important skills young engineers should know entering the workforce?

"Students need more practical 'hands on' experience. I completed an apprenticeship as a machinist before going back to school. This has truly helped me advance past others in my field."

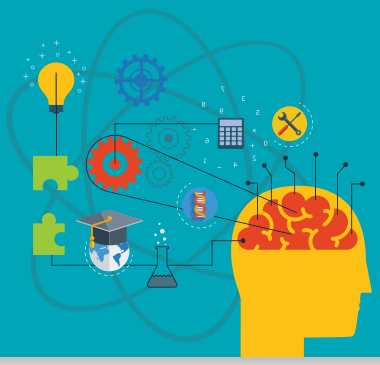
"Students should learn how to troubleshoot the systems they are involved with, and how to simplify the maintenance and repair of the products they design."

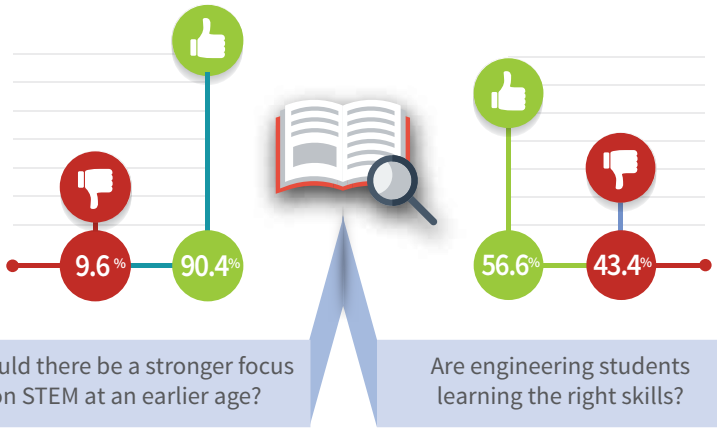
Which certifications do you have?



Certified Drafter	10%
Certified Engineering Technician	11.%
European Engineer by European Federation of National Engineering Associations	0.9%
International Council on Systems Engineering Certification (Systems Engineering)	1.5%
Member of the National Academy of Engineering	1.5%
National Institute For Certification In Engineering Technologies	1.2%
Professional Engineer	29.7%
Senior Member of the Institute of Electrical and Electronics Engineers	1.5%
Senior Member of the Institute of Industrial Engineers	1.5%
Six Sigma by Institute of Industrial Engineers	5.6%
Structural Engineer	1.8%
Structural Engineering Certification Board	0
NICET Certified Engineering Technologist	0.6%
NICET Associate Engineering Technologist	0.3%
IFPS Certified Fluid Power Engineer	7.1%
IFPS Certified Fluid Power System Designer	2.4%
IFPS Certified Fluid Power Specialist	22.6%
IFPS Certified Fluid Power Technician	7.1%
IFPS Certified Fluid Power Mechanic	4.4%
IFPS Certified Fluid Power Connector & Conductor	2.6%
Other (please specify) View Responses »	31.2%

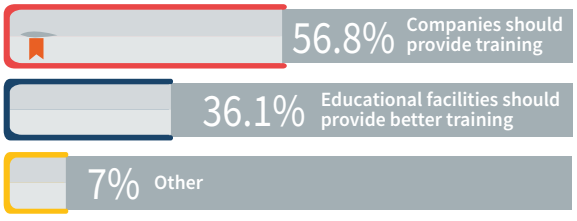
STEM AND EDUCATION





SHOULD COMPANIES PROVIDE MORE SKILL-LEVEL TRAINING TO ENTRY-LEVEL ENGINEERS, OR SHOULD THEY ALREADY HAVE THOSE SKILLS LEARNED?







What kind of early STEM education do you think is needed the most?

Mathematics	28.3%
Shop Class	34.7%
Physics	16%
Chemistry	0.9%
Computer Skills/Coding	12.9%



DOES YOUR COMPANY OFFER STUDENT INTERNSHIPS?

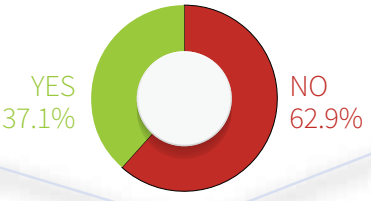
YES 49.4% NO 50.6%

TRAINING + SKILLS

Does your company use training consultants to train engineers before they start on the job?



Does your company have a mentoring program?



Do you find workers right out of school bring new skills to your company?



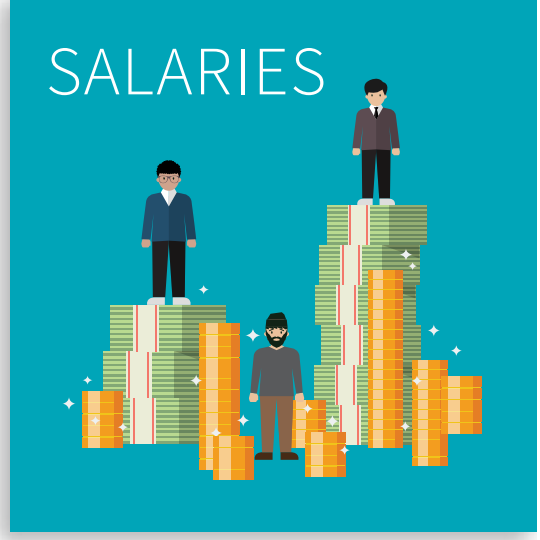
If yes, which skills are they bringing?

- MODELING 46.3%
- 3D PRINTING 27.2%
- ROBOTICS PROGRAMMING 26.1%
- CAD/CAM & SIMULATION 65%

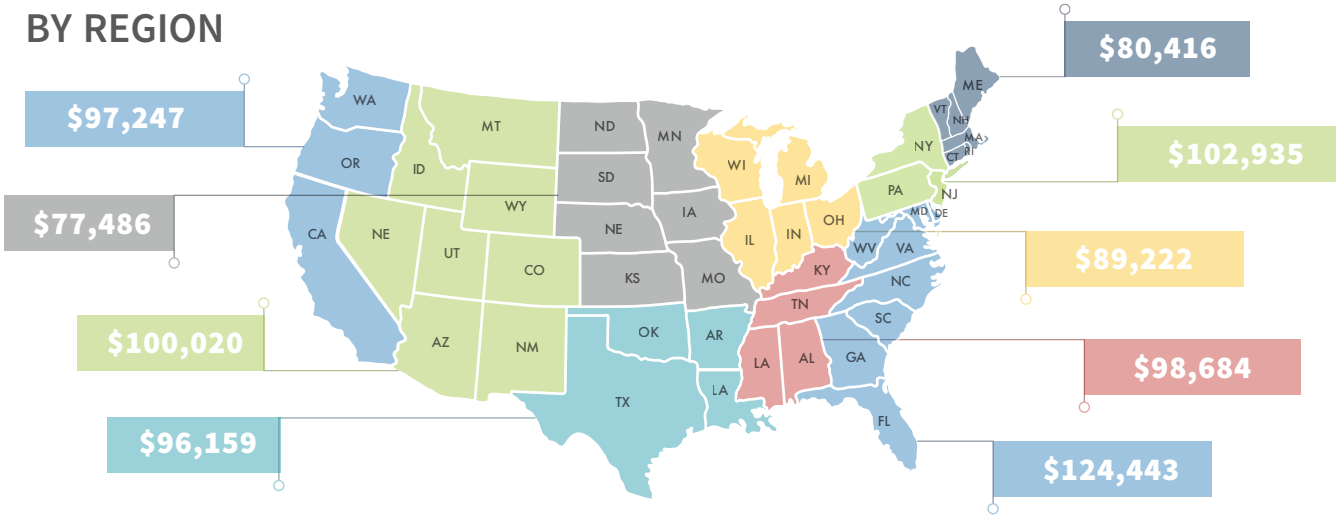


BY INDUSTRY

$132,958.33	Avionics/marine/space
$111,815.79	Research & development
$104,416.67	Software
$98,551.02	Industrial controls systems/equipment
$96,464.79	Mobile Equipment
$88,447.37	Consultant
$88,120.00	Components and subassemblies
$86,750.00	Government/military
$85,420.29	Maintenance & Servicing
$83,692.31	Test and measurement equipment
$82,285.71	Drillng & Mining
$81,857.14	Contract design or manufacturing
$80,563.83	Machine/tool automation



BY REGION





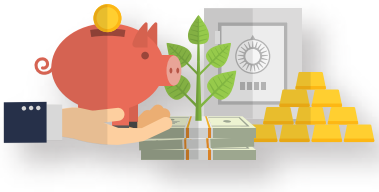
BY EXPERIENCE

40 years or more	$85,518.02
35-39 years	$109,042.68
30-34 years	$100,461.96
25-29 years	$103,132.53
20-24 years	$100,717.11
15-19 years	$90,372.55
10-14 years	$93,616.28
5-9 years	$73,404.76
1-4 years	$60,931.03
Less than 1 year	$70,361.11



BY JOB

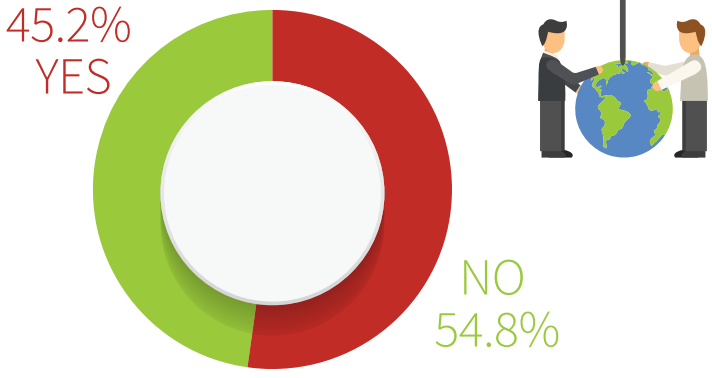
Executive/operating management	$122,661.29
Engineering management	$121,813.56
Design & development engineering	$91,758.20
Other	$72,244.76

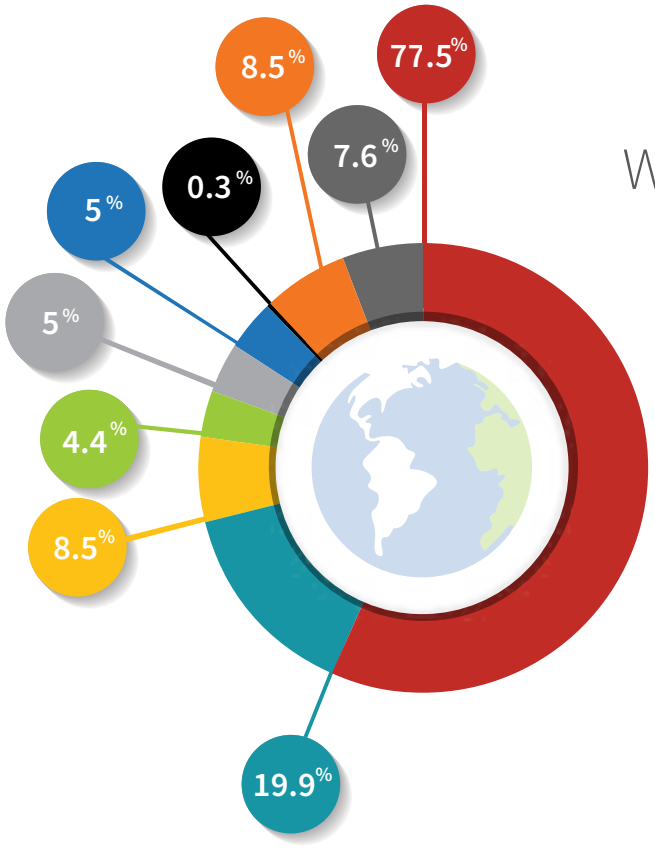


OUTSOURCING



DOES YOUR COMPANY OUTSOURCE?

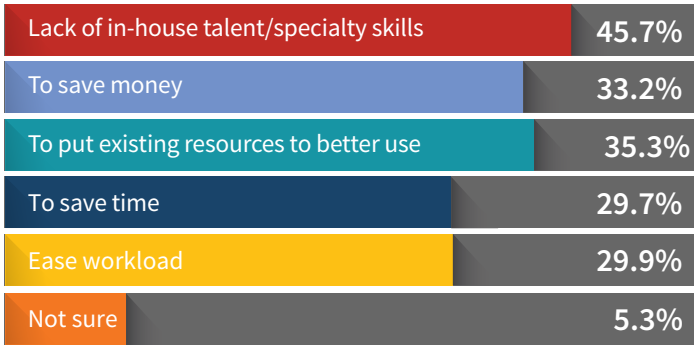




WHERE ARE JOBS GOING?

- UNITED STATES
- CANADA
- CHINA
- INDIA
- MEXICO
- SOUTH AMERICA
- EUROPE
- PACIFIC RIM

REASONS FOR OUTSOURCING



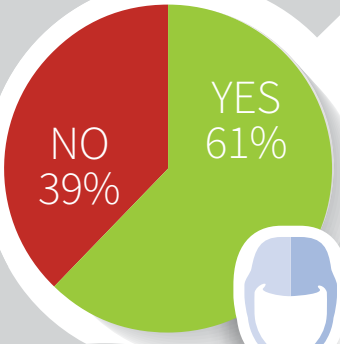
WORK BEING OUTSOURCED

Design	14
Manufacturing/assembly	35.1
Software engineering/development	28
CAD/CAE	29.3
Drafting	31.1
R&D	14
PCB layout	3.4
Design verification	18.1
Software verification/test	9.1
Final test	7
Incoming inspection	3.7

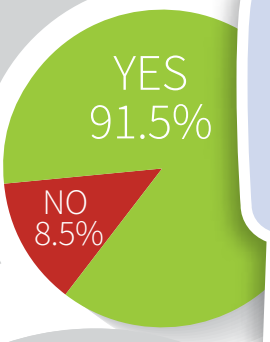
ENGINEERING LABOR FORCE



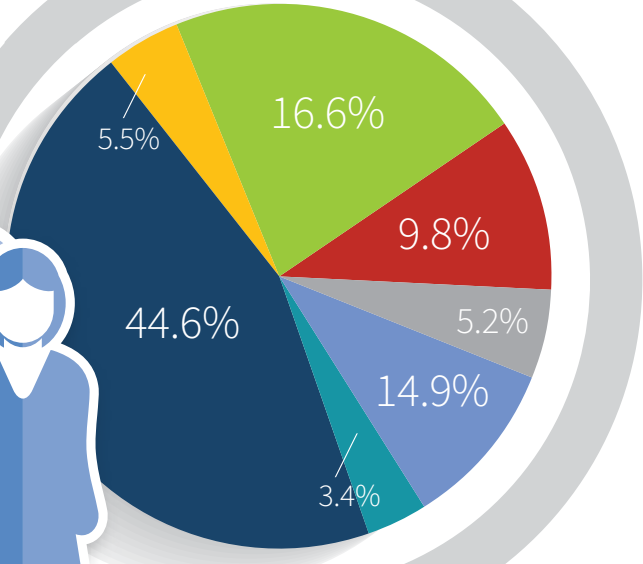
Is there an engineering shortage?



Are we lacking skilled laborers (i.e. welders, machinists, technicians, maintenance, installation)?

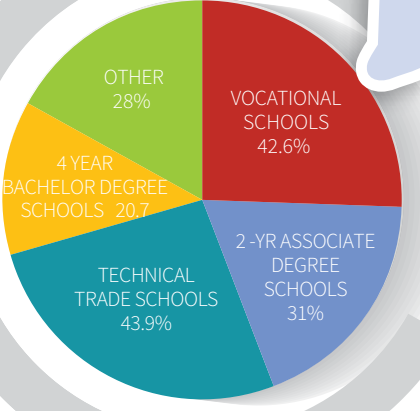






Where do you see the largest job growth potential for the next 5 to 10 years?

- AUTOMATION
- DATA GATHERING + ANALYSIS
- OTHER
- MACHINE MONITORING + CONTROLS
- GAS AND OIL
- AUTOMOTIVE
- CONSTRUCTION



From where do you hire your skilled labor force?

JOB SATISFACTION



HOW SATISFIED ARE YOU IN YOUR JOB?

27.8%	Extremely Satisfied
33.3%	Very Satisfied
31.0%	Satisfied
6.3%	Not Very Satisfied
1.6%	Not At All Satisfied

MOST IMPORTANT FACTORS IN JOB SATISFACTION

7.78	Challenges that accompany the design of new products
7.70	Researching potential design solutions
7.39	Opportunity to design products that can benefit society
7.56	Compensation you receive for the work you do
6.78	Working in team situations with peers
6.87	The pressures associated with solving design problems
6.73	Working independently of others
6.35	The recognition you get from others for the work you do

[Based on a scale of 1 to 10]

REASONS ENGINEERS WOULD LEAVE THE PROFESSION

Try something different	34.3%
Pursue other interests or opportunities	24.8%
Have more freedom/free time	22.5%
Do something more fulfilling or satisfying	21.3%
Do something less stressful	21.3%
Ready to retire	22.2%
To make more money	21.0%
Start a business	18.4%
Burnout	20.6%
No further chance for advancement	14.9%
Cut back on long hours	11.7%
The poor job outlook for engineers	8.9%
Switch to teaching	6.7%

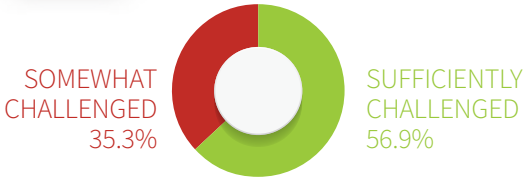


Recommend Engineering?





Feeling Challenged?



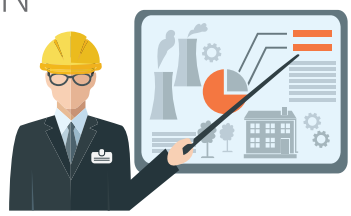
WHAT'S KEEPING ENGINEERS UP AT NIGHT?

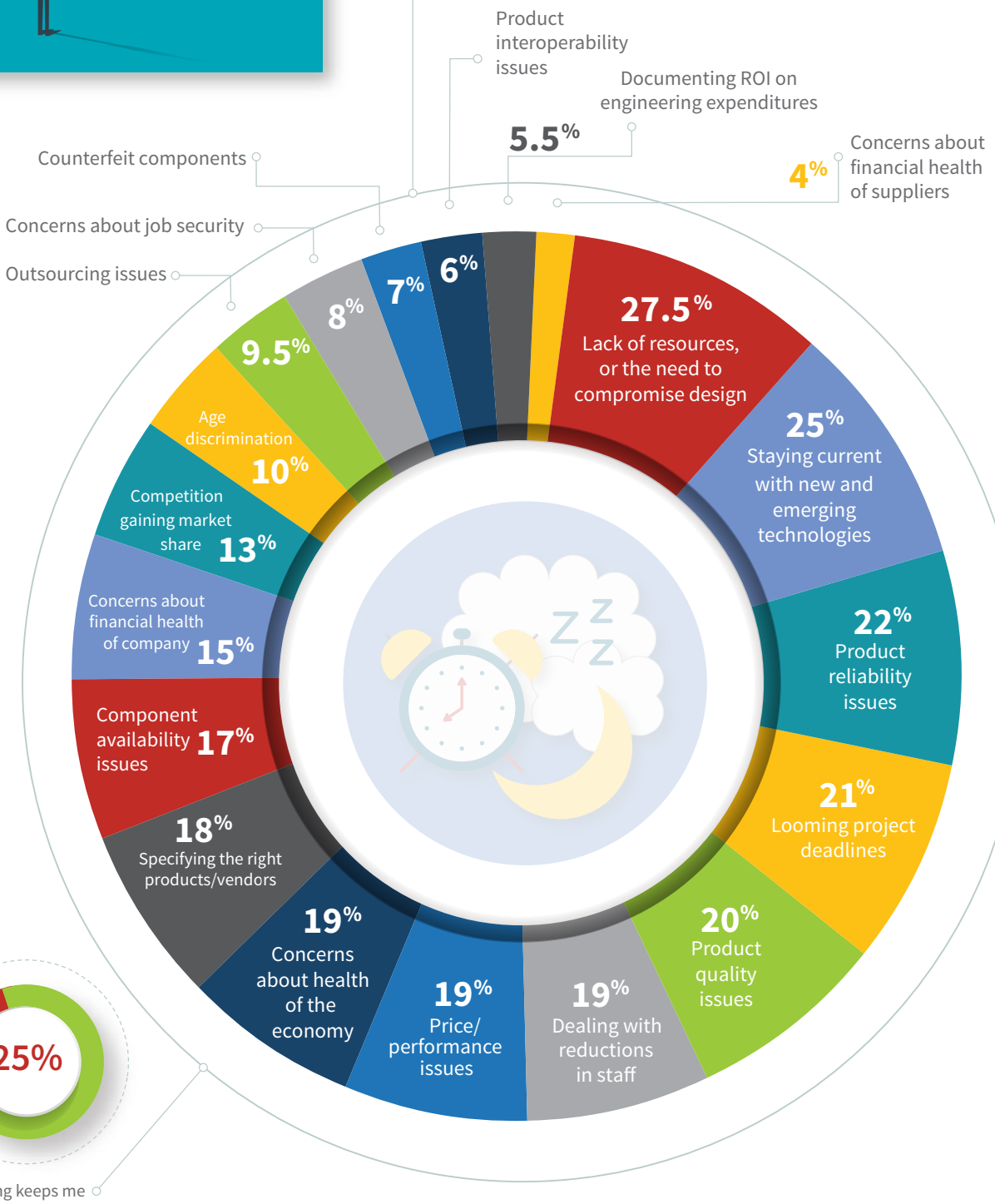


TOP CONCERN

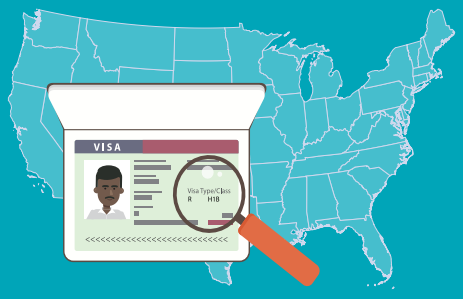
27.5%

Report that a lack of resources, or the need to compromise design is their top concern





ENGINEERING AND SOCIETY



DOES YOUR COMPANY HIRE EMPLOYEES ON H1-B VISAS?

YES 21.6%

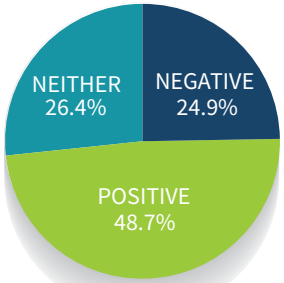
NO 78.4%



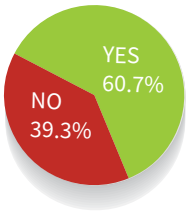
DO YOU THINK H1-B HURTS EMPLOYMENT OPPORTUNITIES FOR ELECTRICAL ENGINEERS IN THE UNITED STATES?

YES 53.1% **NO 46.9%**

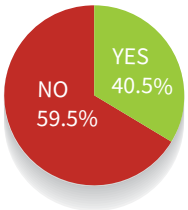
Do you have a negative or positive outlook on President Trump's trade policy, including his decision to withdraw from the Trans-Pacific Partnership and renegotiate the North American Free Trade Agreement.

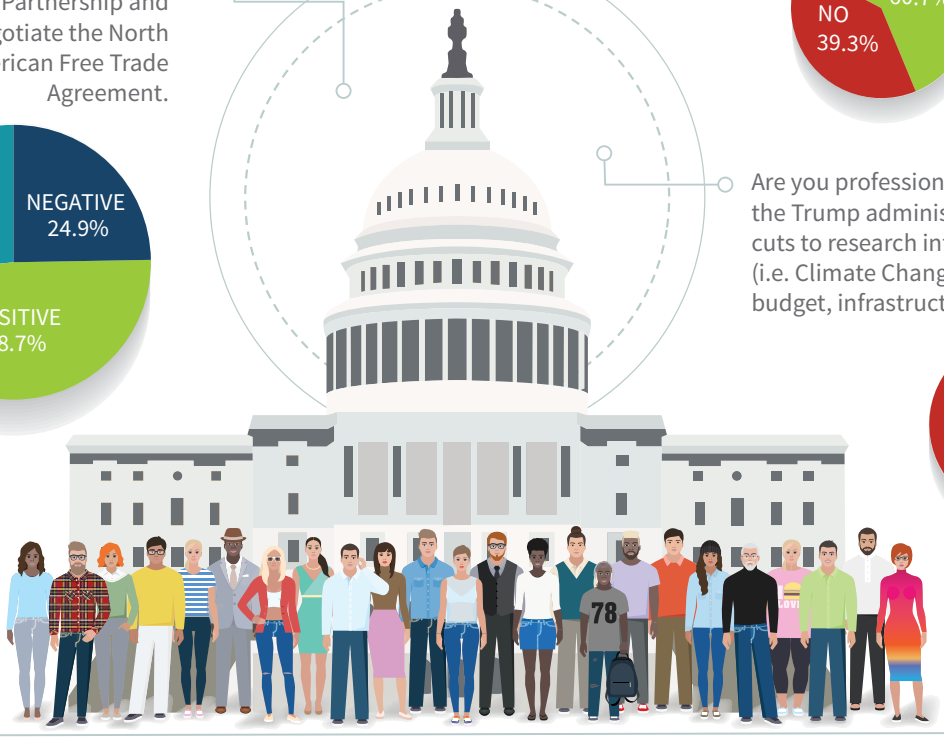


Would you support measures from the Trump administration and Congress to reform how the H1-B system works?



Are you professionally worried about the Trump administration's proposed cuts to research into health and energy (i.e. Climate Change research, NASA budget, infrastructure spending, etc.)





DOES YOUR COMPANY TRACK EMPLOYEE DIVERSITY IN EITHER A PUBLICLY RELEASED OR INTERNAL REPORT?

YES 28.7%

NO 71.3%

40.5% YES

59.5% NO

DO YOU THINK IT IS IMPORTANT TO HAVE A DIVERSE ENGINEERING WORKFORCE, WITH MORE WOMEN AND MINORITY ENGINEERS?